



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Department of Artificial Intelligence and Data Science

Subject Name	Theory of Computations	A.Y	2026-27
Subject Code	303105306	Semester	5
Unit-1			
Sr No	Question	COs	B.T
1	Define alphabet and string with examples.	1	1
2	Explain formal languages and Kleene closure.	1	2
3	Differentiate finite and infinite languages.	1	4
4	Define grammar and its components.	1	1
5	Explain production rules with examples.	1	2
6	Describe derivation in formal grammars.	3	2
7	Differentiate terminals and non-terminals.	3	4
8	Explain leftmost and rightmost derivation.	3	2
9	Construct derivation for a given grammar.	3	3
10	Define Chomsky hierarchy.	1	1
11	Compare Type-0 and Type-1 grammars.	3	4
12	Compare Type-2 and Type-3 grammars.	3	4
13	Explain unrestricted grammars.	4	2
14	Describe context-sensitive grammars.	4	2
15	Explain context-free grammars.	3	2
16	Explain regular grammars with examples.	2	2
17	Classify grammars in Chomsky hierarchy.	3	3
18	Explain applications of formal languages.	1	2
19	Construct grammar for simple language.	3	3
20	Analyze ambiguity in grammar.	3	4
21	Evaluate grammar type for productions.	3	5
22	Design grammar for binary strings.	3	6
23	Construct parse tree for derivation.	3	3
24	Explain relation between grammar and automata.	2	2
25	Create comparison table for Chomsky hierarchy.	3	6



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Unit-2			
Sr No	Question	COs	B.T
1	Define regular expressions and regular languages.	2	1
2	Explain DFA with example.	2	2
3	Construct DFA for strings ending with 01.	2	3
4	Differentiate DFA and NFA.	2	4
5	Convert NFA to DFA.	2	3
6	Explain equivalence of DFA and NFA.	2	2
7	Define Moore machine with example.	2	1
8	Define Mealy machine with example.	2	1
9	Compare Moore and Mealy machines.	2	4
10	Convert Mealy to Moore machine.	2	3
11	Convert Moore to Mealy machine.	2	3
12	Explain regular grammar with examples.	2	2
13	Convert regular grammar to FA.	2	3
14	Convert FA to regular grammar.	2	3
15	Explain closure properties of RL.	2	2
16	Explain union and intersection of RL.	2	2
17	Describe pumping lemma for RL.	2	2
18	Test regularity using pumping lemma.	2	5
19	Construct RE for given language.	2	3
20	Minimize given DFA.	2	3
21	Analyze equivalent states in DFA.	2	4
22	Design DFA for binary divisible by 3.	2	6
23	Evaluate DFA and NFA complexity.	2	5
24	Create FA for identifier validation.	2	6
25	Explain applications of FA.	2	2



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Unit-3			
Sr No	Question	COs	B.T
1	Define CFG and CFL with examples.	1	1
2	Explain parse tree in CFG.	3	2
3	Construct parse tree for expressions.	3	3
4	Explain ambiguity in CFG.	3	2
5	Analyze ambiguous grammar.	3	4
6	Define Chomsky Normal Form.	3	1
7	Convert CFG into CNF.	3	3
8	Explain Greibach Normal Form.	3	2
9	Define Pushdown Automata.	3	1
10	Explain working of PDA.	3	2
11	Construct PDA for $a^n b^n$.	3	3
12	Explain equivalence of PDA and CFG.	3	2
13	Differentiate DPDA and NPDA.	3	4
14	Explain closure properties of CFL.	3	2
15	Explain pumping lemma for CFL.	5	2
16	Test CFL using pumping lemma.	5	5
17	Construct CFG for palindrome language.	3	3
18	Analyze derivation in CFG.	3	4
19	Explain deterministic PDA.	3	2
20	Define context-sensitive grammar.	4	1
21	Compare CFL and CSL.	4	4
22	Design CFG for balanced parentheses.	3	6
23	Evaluate ambiguity in grammar.	3	5
24	Create PDA for palindrome recognition.	3	6
25	Explain applications of CFG and PDA.	3	2



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Subject Name	Theory of Computations	A.Y	2026-27
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Unit-4			
Sr No	Question	COs	B.T
1	Define Turing Machine with example.	4	1
2	Explain components of TM.	4	2
3	Describe transition function in TM.	4	2
4	Construct TM for $a^n b^n$.	4	3
5	Construct TM for palindrome recognition.	4	3
6	Explain recursively enumerable languages.	4	2
7	Explain recursive languages.	4	2
8	Differentiate recursive and RE languages.	5	4
9	Explain closure properties of recursive languages.	5	2
10	Explain closure properties of RE languages.	5	2
11	Describe variants of TM.	4	2
12	Compare multitape and single tape TM.	4	4
13	Explain nondeterministic TM.	4	2
14	Explain equivalence of DTM and NTM.	4	2
15	Define unrestricted grammar.	4	1
16	Explain equivalence of TM and UG.	4	2
17	Describe TM as enumerator.	4	2
18	Construct TM enumerator for even numbers.	4	3
19	Analyze acceptance and rejection in TM.	4	4
20	Evaluate computational power of TM.	4	5
21	Design TM for binary addition.	4	6
22	Create state diagram for TM.	4	6
23	Analyze language accepted by TM.	4	4
24	Evaluate variants of TM.	4	5
25	Explain applications of TM.	4	2



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Unit-5			
Sr No	Question	COs	B.T
1	Define Church-Turing Thesis.	5	1
2	Explain significance of Church-Turing Thesis.	5	2
3	Define Universal Turing Machine.	4	1
4	Explain working of UTM.	4	2
5	Illustrate simulation using UTM.	4	3
6	Define universal language.	5	1
7	Explain diagonalization language.	5	2
8	Analyze diagonalization method.	5	4
9	Explain undecidable problems.	5	2
10	Differentiate decidable and undecidable problems.	5	4
11	Explain halting problem.	5	2
12	Analyze why halting problem is undecidable.	5	4
13	Evaluate role of diagonalization.	5	5
14	Explain reducibility in TOC.	5	2
15	Discuss applications of undecidability.	5	2
16	Analyze recursive and RE languages.	5	4
17	Evaluate UTM in computability.	5	5
18	Design flowchart for TM simulation.	4	6
19	Create model for undecidable language.	5	6
20	Explain acceptance problem for TM.	5	2
21	Evaluate Church-Turing Thesis critically.	5	5
22	Analyze computational limits of machines.	5	4
23	Construct relation between UTM and undecidability.	5	6
24	Explain diagonalization proof with example.	5	3
25	Create comparison table for decidable and undecidable problems.	5	6